

Chapter 5: Mathematics Logic

Class 21 – Logical Reasoning, Visual Puzzles & Brain Teasers

Learning Objectives

After completing this lesson, students will be able to:

- Develop logical thinking and reasoning skills.
 - Solve simple visual puzzles and brain teasers.
 - Identify patterns and relationships between shapes and pictures.
 - Improve observation and analytical thinking.
 - Solve analogy and odd-one-out questions.
 - Apply logical reasoning to everyday situations.
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Introduction

Logical reasoning helps us think carefully before making decisions. It improves our ability to observe, compare, analyze, and solve problems. Visual puzzles and brain teasers make learning mathematics interesting and enjoyable while strengthening memory and concentration.

In this lesson, students will solve different types of reasoning puzzles using pictures, shapes, numbers, and logical thinking.

What is Logical Reasoning?

Logical reasoning is the ability to think carefully and solve problems using facts, observation, and reasoning.

Importance of Logical Reasoning

- Improves thinking skills.
- Develops problem-solving ability.
- Strengthens observation.
- Helps in making correct decisions.
- Builds confidence in mathematics.

Picture-Based Reasoning

Students observe pictures carefully and answer questions.

Example



Question: Which fruit is different?

Answer: 🍌 Banana

Another Example



Question: Which animal is different?

Answer: 🐱 Cat

Visual Pattern Recognition

Students identify the next picture or shape in a pattern.

Example 1



Answer: ■

Example 2



Answer: ●

Example 3



Answer: 

Find the Odd One Out

Choose the object that does not belong to the group.

Examples

1. Apple – Mango – Banana – Car

Answer: Car

2. Circle – Square – Triangle – Pencil

Answer: Pencil

3. Bus – Train – Bicycle – Banana

Answer: Banana

Shape Matching

Match each shape with the correct shadow or identical shape.

Examples

Shape	Match
Circle	○
Triangle	△
Square	□
Rectangle	▭
Star	★

Mirror Images (Basic)

Students learn how objects look in a mirror.

Examples

- Left becomes Right.
- Right becomes Left.

Practice

If a letter **B** is placed before a mirror, students identify its mirror image.

Observe simple shapes and identify the correct mirror image.

Complete the Pattern

Complete the missing picture or number.

Examples

1. ▲ ▲ ● ▲ ▲ ● ____

Answer: ▲

2. 2, 4, 6, 8, ____

Answer: 10

3. 5, 10, 15, ____

Answer: 20

Direction Sense (Basic)

Students identify directions using simple situations.

Directions

- North
- South
- East
- West
- Left
- Right

Example

A student walks **5 steps forward**, turns **right**, and walks **3 steps**.

Question: Which direction did the student turn?

Answer: Right

Analogy (Simple Comparisons)

Students identify relationships between two objects.

Examples

Teacher : School :: Doctor : **Hospital**

Bird : Nest :: Bee : **Hive**

Book : Read :: Food : **Eat**

Pen : Write :: Brush : **Paint**

Brain Teasers

Students solve simple thinking questions.

Examples

1. Which number comes after 49?

Answer: 50

2. Which number is greater?

35 or 53

Answer: 53

3. I have four legs but cannot walk. What am I?

Answer: A Table

4. Which month has 28 days?

Answer: All months

5. What has hands but cannot clap?

Answer: A Clock

Observation Skills

Students carefully observe pictures and answer questions.

Observation Activity

Look at a classroom picture and identify:

- Number of students
- Number of chairs
- Number of windows
- Number of computers
- Number of books

Critical Thinking Activities

Students solve simple real-life situations.

Examples

- Why should we stand in a queue?
- Why should we wear a helmet while riding a bicycle?
- What should you do if you find a lost school bag?
- Why is teamwork important?

Revision

Students revise:

- Logical Reasoning
- Picture Puzzles
- Shape Patterns
- Odd One Out
- Mirror Images
- Direction Sense
- Analogies
- Brain Teasers
- Observation Skills

Did You Know?

- Logical reasoning helps scientists, engineers, doctors, and programmers solve complex problems.
 - Puzzle games improve memory and concentration.
 - Observation skills help us notice details that others may miss.
 - Brain teasers are a fun way to exercise the brain.
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Points to Remember

- Observe carefully before answering.
- Look for patterns and similarities.
- Think step by step.
- Read every question carefully.
- Use logical thinking instead of guessing.
- Practice puzzles regularly to improve reasoning skills.